AssistAnt Orchestrator Routing Technical Report

Angeles University Foundation · PTF60 Project · BS Computer Science 4

Request	Write a technical summary report on how AssistAnt's Orchestrator routes work to its agents and models, including the fallback chain.
Skill	Technical report
Routing	hard task, best tier
Models	Kilo (keyless)
Review	1 round(s), approved, 59/59 claims grounded
Generated	2026-11-05 23:47

Summary

AssistAnt is a personal agent built as a Next.js website and installable app that wraps up the day in Notion and plans tomorrow [S1]. Every user request enters through the Orchestrator, which matches it against routines—markdown files with case-insensitive regular-expression triggers—and runs the first matching routine's flow [S1][S3]. If no routine matches, the Orchestrator handles the request itself via the unplanned flow and offers to save a new routine [S1][S4]. The Orchestrator distributes work to eleven subagents, each assigned a tier—code, smart, or best—that determines which model chain it queries [S1][S2]. Code-tier agents never call a model and are instant and free, while smart- and best-tier agents request a chain of providers from the model router, walking the chain until an available provider is found [S1][S5]. If every provider in a chain fails or is unavailable, the agent falls back to plain rules, and every flow is written to finish without a model [S9]. In public demo mode, only keyless gateways are offered, and when those are busy, flows fall back to plain rules [S7][S9].

Key findings

- The Orchestrator is the single entry point for all requests and either runs a matched routine or plans the request itself [S1].
- There are eleven subagents: Reader, Summarizer, Matcher, Notes Scout, Chore Keeper, Planner, Writer, Checker, Grocery Clerk, Briefer, and Portfolio Clerk [S1][S2].
- Code-tier agents (Reader, Matcher, Chore Keeper, Planner, Writer, Checker, Grocery Clerk, Portfolio Clerk) never call a model and are instant and free [S1][S2].
- Smart-tier agents are the Summarizer and Briefer; the Notes Scout is the only best-tier agent [S2].
- Each agent asks the model router for its tier—fast, smart, or best—and the router walks that tier's chain of providers, using the first one that passes health checks and is available [S5].
- The best chain is Claude Code (user Pro/Max), Groq gpt-oss-120b, free providers from awesome-free-llm-apis, OpenRouter, then local Ollama [S5].
- The smart chain is Gemini Flash, Groq gpt-oss-120b, free list, OpenRouter, Ollama, then paid Claude API (off by default) [S5].
- The fast chain is Groq gpt-oss-20b, Gemini Flash-Lite, Groq Qwen, free list, OpenRouter, Ollama, then Claude API [S5].
- Provider health statuses are ok, full (daily cap reached), off (switched off in Sources), nokey (no API key set), or offline (Ollama or Claude Code not running); only ok providers are called [S5].

- A 429 response marks a provider exhausted for the rest of the day [S9].
- Three keyless gateways—Kilo Code, LLM7.io, and OVHcloud AI Endpoints—work without an API key but are off by default because they may log prompts [S9].
- All agents read and change data through a single Workspace interface, so flows never call Notion or Sheets directly [S10].
- Routines are markdown files in the routines folder with front matter containing a name, a flow, and case-insensitive regular-expression triggers [S3].
- Built-in routines include Finish day (nightly wrap-up at 22:10), Morning brief (07:00), What's due, Groceries, Add a task, and Portfolio [S4].
- The scheduler checks the clock every 20 seconds and runs each job once per day [S4].
- When no routine matches, the Orchestrator drafts a reusable routine and asks for approval before saving it [S4].
- Setting PUBLIC_DEMO=1 turns AssistAnt into a public demo: the workspace is always the fake sample workspace, only keyless model gateways are offered switched on, and no API key is ever sent [S7].
- Visitor isolation in demo mode is achieved via a random visitor ID cookie set by middleware before the first API call [S7].

How it works

- **Request intake.** Every request goes to the Orchestrator [S1]. The Orchestrator tests the request against every routine's case-insensitive regular-expression triggers and runs the first routine that matches [S3]. If no routine matches, the Orchestrator answers the request using the unplanned flow and drafts a reusable routine, asking for approval before saving it [S1][S4].
- **Agent assignment.** The Orchestrator hands work to one or more of its eleven subagents, each with a tier that decides which line of models it asks for [S1][S2]. Code-tier agents (Reader, Matcher, Chore Keeper, Planner, Writer, Checker, Grocery Clerk, Portfolio Clerk) never call a model, so they are instant and free [S1][S2]. Smart-tier agents (Summarizer, Briefer) and the best-tier agent (Notes Scout) do call the model router [S2].
- **Workspace access.** All agents read and change data through a single Workspace interface covering the day blocks on the Tasks page, Reminders, Notes, the Grocery List, and the portfolio snapshot [S10]. Flows never call Notion or Google Sheets directly [S10].
- **Model routing.** Each agent that needs a model asks the router for its tier—fast, smart, or best [S5]. The router walks that tier's chain of providers in order and uses the first one that is available [S5]. Before calling a provider, the router checks its health; only providers with status "ok" are called [S5]. A provider marked "full" has reached its daily cap, "off" means switched off in Sources, "nokey" means no API key is set, and "offline" means Ollama or Claude Code is not running [S5].
- **Provider chains.**
- **Best:** Claude Code (user Pro/Max) Groq gpt-oss-120b free providers from awesome-free-llm-apis OpenRouter local Ollama [S5].
- **Smart:** Gemini Flash Groq gpt-oss-120b free list OpenRouter Ollama paid Claude API (off by default) [S5].
- **Fast:** Groq gpt-oss-20b Gemini Flash-Lite Groq Qwen free list OpenRouter Ollama Claude API [S5].

- **Fallback to plain rules.** If every provider in a chain fails or is unavailable, the router returns nothing and the agent falls back to plain rules [S9]. Every flow is written so it can finish without a model, keeping AssistAnt working offline and for free [S9].
- **Keyless gateways in demo mode.** In public demo mode (PUBLIC_DEMO=1), only the three keyless gateways—Kilo Code, LLM7.io, and OVHcloud AI Endpoints—are offered, switched on, and no API key is ever sent [S7][S9]. When these gateways are busy, flows fall back to plain rules, the same mechanism used when all model providers in a chain are unavailable [S7][S9].
- **Scheduled runs.** Built-in routines like Finish day (22:10) and Morning brief (07:00) run on a scheduler that checks the clock every 20 seconds and runs each job once per day in the configured time zone [S4]. The scheduler is off in demo mode [S7].
- **Run events and approvals.** Each request is a run that emits events—plan, messages between agents, work done, and the result—streamed live to the browser via Server-Sent Events and replayable later [S6]. Risky changes wait for user approval; the Writer holds new guessed Reminders and the deletion of today's block until the user approves them [S6]. Scheduled runs keep their approvals waiting until the app is opened [S6].
- **Chat and Knowledge Base.** Questions from chat go to the Knowledge Base, where the top passages are retrieved using BM25 keyword ranking (k1=1.4, b=0.75) and the answer cites them as numbered sources [S8]. Requests to do something go to the crew as a run; if the crew is already busy, the chat says so instead of starting a second run [S8].

Risks and limitations

- **Model availability dependency.** If all providers in a tier's chain are unavailable, the agent falls back to plain rules, which may produce less accurate or complete results than a model would [S9].
- **Daily caps.** Providers with daily caps (Gemini Flash at 20 requests/day, Groq at 1,000 requests/day) are skipped once reached, and a 429 response marks the provider exhausted for the rest of the day [S9].
- **Keyless gateway privacy risk.** The three keyless gateways (Kilo Code, LLM7.io, OVHcloud AI Endpoints) may log prompts, which is why they are off by default and only switched on in public demo mode with fake data [S9].
- **Demo mode constraints.** In public demo mode, the workspace is always the fake sample workspace, the scheduler is off, and runs and chat are rate-limited per visitor and per IP address [S7].
- **Offline limitation.** While every flow is written to finish without a model, features that depend on model output (such as summaries, briefs, and deadline extraction) will not function when no provider is available [S9].
- **Approval delays.** Risky changes and scheduled runs wait for user approval, which can delay execution until the app is opened [S6].
- **Chat concurrency.** Only one run can execute at a time; if the crew is busy, new requests from chat are rejected with a message rather than queued [S8].
- **Keyless gateway default-off.** Because keyless gateways are off by default, users without any configured API key or running local model have no model access unless they manually enable them [S9].

Sources

- [S1] architecture.md, part 1
- [S2] architecture.md, part 2
- [S3] routines.md, part 1
- [S4] routines.md, part 3
- [S5] model-router.md, part 1
- [S6] architecture.md, part 3
- [S7] public-demo.md, part 1
- [S8] knowledge-base.md, part 1
- [S9] model-router.md, part 2
- [S10] workspace.md, part 1

How the team worked

You → **Orchestrator** · New request
From now on, always end reports with a 'Risks and limitations' section. Write a technical summary report on how AssistAnt's Orchestrator routes work to its agents and models, including the fallback chain.

Orchestrator → **Memory** · Save standing rule
From now on, always end reports with a 'Risks and limitations' section.

Orchestrator → **Reader** · Assign the crew
Reader Notes Scout Writer Checker Publisher

Reader → **Notes Scout** · Notes to verify
15 notes from 5 files

Notes Scout → **Writer** · Evidence pack
10 passages, S1–S10

Writer → **Checker** · Draft for review
round 1

Checker → **Publisher** · Approved
approved

Publisher → **You** · Report ready
1510 words, 10 sources